

FortiGate Login Failure Detection with MISP Threat Intelligence Enrichment

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Date: February - March 2026

1. Purpose

The purpose of this SOP is to automate the detection, enrichment, and notification of **FortiGate administrative login failure events** using:

- **Wazuh SIEM**
- **n8n SOAR automation**
- **MISP Threat Intelligence**
- **Email Alerting**

This automation helps the Security Operations Center (SOC) quickly determine whether the source IP of a login attempt is **malicious or benign** based on threat intelligence data.

2. Scope

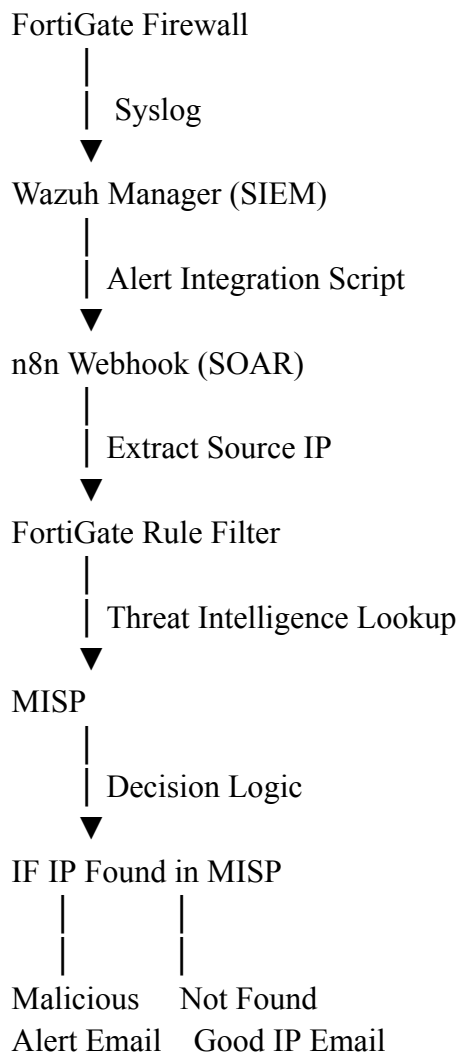
This SOP applies to:

- FortiGate firewall authentication logs
 - Wazuh SIEM alert processing
 - Threat intelligence enrichment using MISP
 - Automated alert notification through email
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3. Systems Involved

System	Purpose
FortiGate Firewall	Generates authentication logs
Wazuh Manager	SIEM platform for log collection and detection
n8n	SOAR automation platform
MISP	Threat intelligence platform
Email Server	Sends security alert notifications

4. Architecture Overview



5. Event Detection

5.1 Event Source

FortiGate firewall generates an event when an administrator login fails.

Example log:

Administrator login failed from https(192.168.0.52)

5.2 Wazuh Rule Detection

Wazuh detects the event using rule:

Rule ID: 110610

Description: FortiGate Admin GUI login failed

Severity Level: 10

Example decoded fields:

Field	Value
srcip	192.168.0.52
action	login
status	failed
reason	invalid user

6. Alert Forwarding

When Wazuh generates an alert, the integration script:

```
/var/ossec/integrations/custom-n8n
```

sends the alert to the n8n webhook.

Example webhook:

```
http://192.168.0.35:5678/webhook/siem-alert
```

The alert is sent as JSON.

7. n8n Workflow Processing

The n8n workflow processes the alert through several stages.

Step 1 — SIEM Webhook

Receives the alert from Wazuh.

Input Example:

```
{  
  "rule": {  
    "id": 110610,  
    "description": "FortiGate: Admin GUI login failed"  
  },  
}
```

```
"data": {  
  "srcip": "192.168.0.52"  
}
```

Step 2 — Extract Source IP

The workflow extracts important fields:

Field	Purpose
srcip	Attacker IP
rule_desc	Alert description
severity	Alert severity
timestamp	Event time

Example expression:

```
{{ $json.body?.data?.srcip || $json.data?.srcip }}
```

Step 3 — Filter Specific FortiGate Rule

The workflow verifies the alert belongs to the correct rule.

Condition:

```
rule_id = 110610
```

This ensures only **FortiGate Admin login failures** trigger enrichment.

Step 4 — MISP Threat Intelligence Lookup

The workflow sends the source IP to the MISP API.

API Endpoint:

```
POST /events/restSearch
```

Request Body:

```
{  
  "value": "192.168.0.52"  
}
```

Purpose:

Check if the IP exists in the **threat intelligence database**.

8. Decision Logic

The workflow evaluates the response from MISP.

Condition:

`response.length > 0`

Case 1 — Malicious IP

If MISP returns:

```
{  
  "response": [  
    {  
      "Event": {  
        "info": "Malware infrastructure"  
      }  
    }  
  ]  
}
```

Then:

`response.length = 1`

Result:

Malicious IP detected

Action:

Send Malicious Alert Email

Case 2 — Good / Unknown IP

If MISP returns:

```
{  
  "response": []  
}
```

Then:

`response.length = 0`

Result:

IP not found in threat intelligence

Action:

Send Good IP Notification

9. Email Alerting

Two types of alerts are generated.

9.1 Malicious IP Alert

Example Email:

Subject: Malicious IP Detected — FortiGate Alert

Contents:

- Source IP
- Rule Description
- Severity
- Timestamp
- Threat Intelligence Result

Example:

Threat Intel: Found in MISP

9.2 Good IP Notification

Example Email:

Subject: Suspicious Login Attempt Detected

Contents:

- Source IP
- Rule
- Severity
- Timestamp

Example message:

Threat Intel: Not found in MISP

10. Analyst Response

SOC analysts should perform the following actions.

If Malicious IP Detected

1. Investigate firewall logs
 2. Verify repeated login attempts
 3. Block IP on firewall if necessary
 4. Create incident ticket
 5. Monitor related activity
-

If IP Not Found

1. Monitor future login attempts
 2. Check if the IP belongs to internal network
 3. Identify potential brute force patterns
-

11. Logging and Monitoring

The following logs should be monitored:

System	Log
Wazuh	/var/ossec/logs/ossec.log
Integrations	/var/ossec/logs/integrations.log
n8n	Workflow executions
MISP	API search logs

12. Security Considerations

- Ensure MISP API authentication is secure
 - Restrict webhook access to trusted systems
 - Monitor abnormal alert volumes
 - Ensure SIEM alert integrity
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13. Future Enhancements

Recommended improvements:

- Add **AbuseIPDB** enrichment
 - Add **VirusTotal** lookup
 - Implement **automatic firewall blocking**
 - Add **brute-force detection**
 - Implement **incident ticketing integration**
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14. Conclusion

This automation enables real-time detection and enrichment of FortiGate login failures by integrating SIEM alerts with threat intelligence. The system reduces manual investigation time and improves SOC response efficiency by automatically identifying potentially malicious IP addresses.
